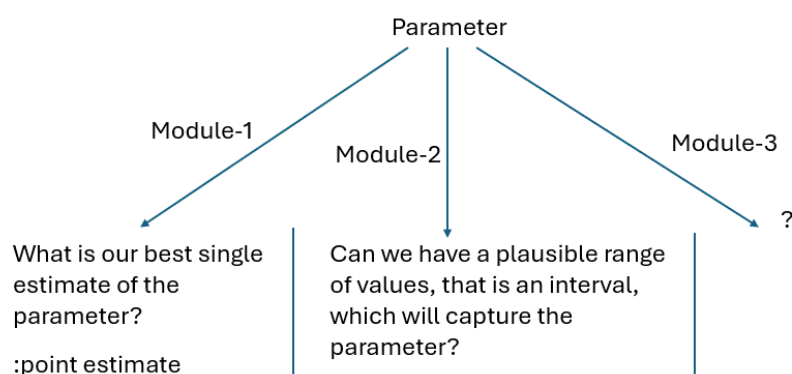


Module -3 part 1: Basic Hypothesis Testing



A motivating example:

Historically, the average score for Section_____ in the first Quiz of Quant course in Wonderful Institute of Management has been 55 (out of 100), with a standard deviation of 5. However, this time the students of that section, after taking the quiz, thought that the average should be better this time. They sent their CR to the faculty to know the average score for their section. The faculty did not reveal the average score to the CR but provided a random sample of 30 scores (anonymously). The average of those 30 scores is 57. What should the CR conclude from this data – is the average performance better this time?

57 is of course higher than 55! Should the CR then announce that the average is better this time? _____

_____.

Hypothesis testing provides a structured way to decide whether _____

_____.

We start with summarizing different steps involved in a hypothesis testing:

Step 1: Forming the null and alternative hypothesis:

Figuring out the null and alternative hypotheses in a given context can sometimes be tricky. The following guidelines can help:

Null Hypothesis (H_0)	Alternative Hypothesis (H_1)
Represents the <i>status quo</i> , baseline, or default assumption.	Represents any change or departure from the default/baseline.
Refers to what has been true historically, or what is currently believed/assumed.	Refers to a new belief, claim, or possibility being tested.
Indicates <i>no difference</i> or <i>no effect</i> .	Indicates a <i>non-zero difference</i> or <i>presence of an effect</i> .
By convention, we assume the null to be true until we find strong evidence against it.	This is the claim we actively investigate and try to gather evidence for.

Rules for Formulating Hypotheses

1. Always include "=", "≥", or "≤" in H_0 , and use "≠", ">", or "<" in H_1 .
 - (This is a must-follow rule. If ignored, technical inconsistencies will arise.)
2. Together, H_0 and H_1 must cover all possible outcomes.
3. In some cases, it may be more natural to frame the alternative hypothesis first.

So, in our case,

Step 2: Test statistic and its distribution under null hypothesis:

Recall the earlier example. The CR knows that the sample average score (for 30 random students) is $\bar{x}$ (which is 57 in this case). The CR has to decide whether the overall section average, that is μ , is higher than the null value μ_0 , that is 55 in this case.

Based on this information, when will the CR tend to believe that the overall section average, that is μ , is higher than the null value μ_0 ? When _____

But how should one decide that?

A _____ helps to decide that!

A test statistic is such a quantity that facilitates decision making in hypothesis testing. In this case, test statistic will be as follows:

_____.

So, in this case, the test statistic being higher and higher implies one will tend

to _____ the null hypothesis more and more.

But how to check whether the test statistic is high enough to reject the null?

Step 3: Decision making

To answer the above question, just check if _____ area of the test statistic under $N(0,1)$ is sufficiently _____, which will automatically imply that the test statistic is high enough!

That area is called the _____.

Thus, smaller p-value will lead to the rejection of the null hypothesis. In practice, we reject the null hypothesis if p-value is smaller than α , known as level of significance. Some commonly used values of α are 0.05, 0.01, 0.1.

In this example, since the alternative is '>' type, we are taking p-value as the right-side area of the observed test statistic. However, in general, p-values should be obtained using the following rules (which are very intuitive!).

The p-value rules (please keep these in mind):

- For the '>' type of alternative, the p-value is the **right-tail area** beyond the test statistic.
- For the '<' type of alternative, the p-value is the **left-tail area** beyond the test statistic.
- For the '≠' type of alternative, the p-value is $2 \times$ (right-tail area beyond the absolute value of the test statistic).

There is another approach of decision making, called “critical value” approach. We will discuss that approach later.

Now, let's complete the exercise.

Test statistic value =

_____.

p-value = _____

= _____

Decision: _____.

One sample test for mean (population variance unknown)

In the previous example, population variance was known. If population variance is unknown, what adjustments need to be made? In that case, test statistic would be

and while computing p-values, one has to consider _____

_____ instead of $N(0,1)$ distribution.

Ex 2: (from the book)

Annual per capita consumption of milk is 21.6 gallons (*Statistical Abstract of the United States: 2006*). Being from the Midwest, you believe milk consumption is higher there and wish to support your opinion. A sample of 16 individuals from the midwestern town of Webster City showed a sample mean annual consumption of 24.1 gallons with a standard deviation of $s = 4.8$.

- a. Develop a hypothesis test that can be used to determine whether the mean annual consumption in Webster City is higher than the national mean.

Ex 3: (from the book)

Joan's Nursery specializes in custom-designed landscaping for residential areas. The estimated labor cost associated with a particular landscaping proposal is based on the number of plantings of trees, shrubs, and so on to be used for the project. For cost estimating purposes, managers use two hours of labor time for the planting of a medium-sized tree. Actual times from a sample of 10 plantings during the past month are as follows (times in hours): 1.7, 1.5, 2.6, 2.2, 2.4, 2.3, 2.6, 3.0, 1.4, 2.3. With a .05 level of significance, test to see whether the mean tree-planting time used by the manager is correct or not.

Ex 4:

A bakery sells loaves of bread with a printed weight of 500 grams. If the average weight of the loaves is found to be less than the printed weight, the bakery may face legal action and be required to pay a penalty. To check this, the quality control officer takes a random sample of 25 loaves, finding a sample mean weight of 497 grams with a sample standard deviation of 6 grams. What should be the conclusion of a hypothesis testing at 5% level of significance?

Ex: 5

A company manufactures ball bearings that are designed to have a diameter of 5 mm. If the average diameter of the supplied ball bearings is less than 5 mm, the supply must be scrapped, and the company will incur a penalty. If the average diameter is greater than 5 mm, the ball bearings must be reworked to bring them back to the design specification.

To check compliance, the company selects a random sample of 50 ball bearings and measures their diameters. The sample yields a mean diameter of 4.92 mm with a sample standard deviation of 0.20 mm. What should be the conclusion of a hypothesis testing at 5% level of significance?

